

ALCONBURY, United Kingdom

WMO: 035620

Lat: 52.367N

Lon: 0.217W

Elev: 48

StdP: 100.75

Time Zone: 0.00 (EUW)

Period: 83-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-4.2	-3.0	-9.0	1.8	-3.9	-7.1	2.1	-1.7	13.6	7.8	12.2	7.7	2.5	70	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	26.8	18.7	24.2	17.5	23.0	16.6	19.5	25.0	18.3	23.3	17.3	21.6	3.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.3	21.5	16.2	11.6	20.5	15.2	10.9	19.4	55.3	24.8	51.8	23.4	48.8	21.9	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 10.5	9.2	8.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-8.0	20.5	3.0	1.2	-10.1	21.4	-11.9	22.1	-13.5	22.8	-15.7	23.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	4.2	3.7	6.4	8.3	11.6	14.2	17.1	16.9	14.1	10.9	6.6	5.6
	DBStd	5.53	3.49	3.67	2.88	2.94	2.76	2.75	2.71	2.84	2.66	3.24	3.10	3.51
	HDD10.0	834	179	177	115	69	14	1	0	0	2	28	107	141
	HDD18.3	3081	437	409	369	302	211	126	58	63	128	232	352	394
	CDD10.0	839	1	1	4	17	62	128	219	214	126	55	6	5
	CDD18.3	44	0	0	0	0	1	3	19	19	2	1	0	0
	CDH23.3	314	0	0	0	0	7	26	138	129	11	3	0	0
	CDH26.7	53	0	0	0	0	0	1	24	26	1	1	0	0
Wind	WSAvg	3.7	4.4	4.2	4.2	4.0	3.6	3.3	3.2	3.1	3.4	3.5	3.6	3.8
Precipitation	PrecAvg	537	44	30	41	43	46	49	45	51	50	45	49	46
	PrecMax	729	88	74	82	94	106	123	121	106	122	132	100	99
	PrecMin	280	8	9	3	7	5	2	7	9	4	2	22	11
	PrecStd	88	18	16	22	22	25	28	26	25	32	32	20	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	14.0	16.8	20.1	24.1	26.1	29.2	29.8	24.9	22.0	16.0	14.1
		MCWB	11.3	11.7	12.0	14.3	16.4	18.1	20.2	19.6	16.9	16.8	13.6	11.6
	2%	DB	11.9	11.9	13.9	17.2	21.1	23.8	26.9	26.8	22.2	18.8	13.8	13.1
		MCWB	10.2	9.8	10.8	12.9	15.5	17.3	18.9	18.5	16.1	15.0	11.7	10.9
	5%	DB	10.8	10.8	12.2	15.1	18.9	21.8	24.8	24.2	20.8	17.0	12.2	12.0
		MCWB	9.0	8.8	9.7	11.6	13.9	15.9	18.0	17.3	15.6	14.0	10.3	9.9
	10%	DB	9.1	9.0	11.0	13.1	17.0	19.9	22.9	22.8	18.9	15.9	11.2	11.0
		MCWB	7.9	7.7	8.7	9.9	12.6	15.0	16.9	16.5	14.5	13.0	9.3	8.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.7	12.8	15.3	17.7	19.2	22.1	20.1	18.3	17.3	13.4	12.5
		MCDB	12.8	13.4	15.4	18.9	22.7	24.3	27.2	27.4	22.0	21.2	15.5	13.8
	2%	WB	10.5	10.2	11.3	13.4	15.9	17.6	20.0	19.1	16.9	15.4	12.2	11.3
		MCDB	11.8	11.6	13.6	16.3	20.3	22.2	24.9	24.9	20.7	18.1	13.8	12.8
	5%	WB	9.1	8.9	10.2	11.9	14.5	16.4	18.7	18.1	16.0	14.2	11.1	10.2
		MCDB	10.3	10.2	11.8	14.9	18.4	20.7	23.4	23.3	19.7	16.6	12.4	12.0
	10%	WB	7.9	7.6	9.2	10.5	13.0	15.3	17.4	17.1	15.2	13.1	9.9	9.1
		MCDB	9.0	8.6	10.8	13.1	16.1	19.4	21.5	21.5	18.2	15.4	11.4	10.9
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.3	6.6	7.5	8.2	8.2	9.0	9.1	7.8	6.6	5.3	4.3
		MCDBR	6.1	6.7	7.9	9.8	12.0	11.5	12.2	11.7	10.4	7.6	6.0	4.9
		MCWBR	5.7	5.6	6.0	6.5	7.1	6.7	6.3	5.7	6.0	5.2	5.2	4.5
	5% WB	MCDBR	6.2	6.4	7.6	9.3	11.1	10.4	11.2	10.7	9.0	6.9	6.0	5.1
		MCWBR	6.1	5.7	6.1	6.4	7.1	6.5	6.4	5.6	6.0	5.1	5.5	4.8
Clear-Sky Solar Irradiance	taub	0.323	0.340	0.373	0.398	0.402	0.405	0.404	0.393	0.372	0.354	0.327	0.313	
	taud	2.369	2.386	2.284	2.227	2.254	2.268	2.282	2.336	2.397	2.435	2.416	2.350	
	Ebn at Noon	667	771	809	834	849	848	842	832	806	744	659	607	
	Edh at Noon	61	79	105	125	128	128	124	111	93	74	58	52	
All-Sky Solar Radiation	RadAvg	0.76	1.44	2.52	3.90	4.78	5.12	4.91	4.08	3.06	1.78	0.93	0.61	
	RadStd	0.07	0.18	0.30	0.42	0.32	0.47	0.52	0.29	0.27	0.16	0.10	0.07	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (90 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page